

Sidharth Vijayan Krishnan

Java Backend & Full-Stack Engineer

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SUMMARY

M.Sc. Computer Science graduate with three years of professional software experience in Java development and test automation. Builds full-stack applications with Java/Spring Boot and React, backed by automated testing and CI/CD pipelines. Based in Hannover and available immediately. English C2, German B1 and working toward B2.

SKILLS

Programming Languages — Java, Python, JavaScript, TypeScript, C++, SQL

Backend Frameworks & Technologies — Spring Boot, Spring Security, Spring Data JPA, Thymeleaf, REST APIs

Frontend Development — React, Angular, HTML5, CSS3, Vite, React Router

Databases — PostgreSQL, MySQL, Microsoft SQL Server

DevOps & Build Tools — Git, Docker, Linux, CI/CD, GitHub Actions, Jenkins, Maven

Data Science & Analytics — Pandas, NumPy, Matplotlib, Jupyter

Software Testing & QA — Selenium WebDriver, Appium, JUnit, TestNG, Vitest, JMeter

Methodologies & Practices — Agile, Scrum, TDD, BDD

EDUCATION

M.Sc. Computer Science <i>Technische Universität Clausthal</i>	01.2021 – 09.2025 <i>Clausthal, Germany</i>
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B.Tech. Computer Science and Engineering <i>Mar Baselios Institute of Technology and Science (MBITS)</i>	08.2011 – 05.2015 <i>Kerala, India</i>
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EXPERIENCE

Test Automation Engineer Intern <i>Rogersoft Technologies Pvt. Ltd.</i>	12.2018 – 03.2019 <i>Kerala, India</i>
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- Designed automated testing frameworks and CI/CD pipelines to accelerate delivery and enforce quality across web and mobile platforms.
- Automated 150+ test cases with Selenium and TestNG, cutting regression testing time by 40%.
- Achieved 85% automated test coverage for core iOS/Android flows using Appium.
- Integrated test suites into Jenkins, GitLab CI, and GitHub Actions, reducing developer feedback loops from days to under 2 hours.
- Load-tested 5,000+ concurrent users in JMeter, identifying and resolving 3 critical performance bottlenecks.
- Streamlined dependencies with Maven, improving build stability and cutting setup time by 30%.

Software Developer & Academic Coordinator <i>Matrix Engineering Pvt. Ltd.</i>	11.2015 – 11.2018 <i>Kerala, India</i>
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- Engineered Java applications (Core Java, JDBC, MySQL) across the full SDLC to automate grading, attendance, and student records.
- Built backend logic and database workflows that streamlined result generation and performance tracking.
- Mentored student projects and delivered training in Core Java, OOP, Data Structures, Algorithms, SQL, and Servlets/JSP.
- Developed technical documentation, coding exercises, and evaluation frameworks for structured software engineering education.
- Coordinated academic operations across five departments.

- Built responsive UI components with HTML5, CSS3, and JavaScript, improving cross-browser compatibility and boosting engagement by 15%.
- Designed and optimized SQL relational database schemas to securely manage 10,000+ user records.
- Resolved 30+ backend and frontend bugs, ensuring a stable production release.

RESEARCH & PROJECTS

Agent-Based Simulation of Emergency Evacuation in University Lecture Halls

Master Thesis (Agent-Based Modelling, Multi-agent System, Netlogo, Python)

- Designed an agent-based simulation analyzing evacuation dynamics across architectural design, human psychology, and crowd flow.
- Built a multi-agent NetLogo simulation modeling 200+ concurrent agents with varied mobility and behavioral strategies.
- Modeled psychological decision-making factors across 5 distinct lecture hall architectures.
- Automated 9,000+ simulation runs via NetLogo BehaviorSpace to parameter-sweep exit configurations and blockage scenarios.
- Engineered Python data pipelines to process and visualize simulation datasets tracking egress time, flow rates, and bottlenecks.

Agent-Oriented Programming: Designing BDI Agents in GOAL for an Exploration and Coordination Problem Based on the MAPC

Master Project (Agent-Oriented Programming, Multi-Agent System, GOAL, Java)

- Engineered autonomous cognitive agents to solve team exploration challenges in a dynamic hex grid.
- Programmed autonomous BDI cognitive agents using GOAL for complex environmental reasoning.
- Developed decentralized algorithms for task distribution and navigation without a central controller.
- Designed communication protocols to share spatial knowledge and optimize collective mapping.
- Formulated logic-based action rules for pathfinding and decision-making under time constraints.

Course Timetable Planner (Full-Stack SPA & REST API)

React 19, TypeScript, Java 17, Spring Boot 3, PostgreSQL, Docker, Vitest, JUnit 5

- Engineered a decoupled REST API and dual clients (React SPA and Thymeleaf web app) for course scheduling with automatic conflict detection.
- Built a single-page frontend using React, Vite, and React Router, featuring a typed API layer and client-side state management.
- Implemented role-based access control using Spring Security, enforcing coordinator and instructor permissions consistently across the SPA and API endpoints.
- Containerized the system with Docker, documented endpoints via OpenAPI, and built dedicated test suites using Vitest, React Testing Library, and JUnit 5.

Quiz Assessment Engine

Java 17, Swing, Jackson, JUnit 5, Maven

- Designed an event-driven quiz engine using four GoF design patterns (Factory, Builder, Strategy, Observer) to decouple authoring, scoring, and tracking.
- Built a Swing desktop GUI with live countdown timer, randomized selection from 120+ self-validating questions, and a session leaderboard.
- Implemented polymorphic JSON persistence with Jackson and achieved full JUnit 5 coverage, including Mockito unit tests and headless Swing tests.

LANGUAGES

English – C2, Fluent

German – B1, Actively working toward B2

Malayalam – Native